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Even Sem End Term Examination-2025

Name of the Course: B.Tech. Biotechnology/Hons

Semester: VIII

Name of the Paper: Omics Technology

Paper Code: TBT-801

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain the key components of the human genome and describe how genome is organized within a nuclei?	CO 1
(b)	Describe how exon-intron trapping works and how it can be used to study gene structure?	
(c)	Analyze the strengths and limitations of 454 Pyrosequencing compared to other next-generation sequencing methods, such as Illumina sequencing.	
Q2	(20 marks)	
(a)	Evaluate which approach—forward or reverse genetics—would be more effective for studying gene function in a specific experimental context, such as disease research?	CO 2
(b)	Describe how RNA interference could be applied to silence a specific gene in a mammalian cell line. What steps would you follow in this experiment?	
(c)	Explain the principle of the CRISPR-CAS9 gene editing technology and how it is used to modify genes in a targeted manner.	
Q3	(20 marks)	
(a)	Comment upon yeast as a model experimental organism in life science research.	CO 3
(b)	Write short notes on any two: a). Epigenetics, b)- Metagenomics, c)- Metabolomics.	
(c)	Analyze the role of <i>Arabidopsis thaliana</i> in understanding plant gene functions.	
Q4	(20 marks)	
(a)	Explain the principle of iso electric focusing (IEF). How IEF gives an edge to 2 D SDS-PAGE when compared with SDS-PAGE?	CO 4
(b)	Describe how you would apply mass spectrometry to identify proteins in a complex cell lysate sample?	
(c)	Design an experiment using a yeast two-hybrid system to study protein-protein interactions.	
Q5	(20 marks)	
(a)	What is transcriptomics? Describe key techniques used in transcriptomic analysis.	CO 5
(b)	Outline how you would use microarrays to compare gene expression profiles between cancerous and normal tissues?	
(c)	Demonstrate how you would use real-time PCR to quantify the expression of a target gene in treated vs. untreated cells?	

